

University of Bahrain

ITCIS714 · Advanced Artificial Intelligence

Assignment I: Critical Exploration of a Recent AI Advancement

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Federated Low-Rank Adaptation for Large Language Model Fine-Tuning

A Critical Review and Empirical Analysis of Aggregation Strategies

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1 Introduction

Large Language Model (LLM) have shown the power in various applications such as chatbots, virtual assistants, and healthcare. However, they require a large computational power to fine-tune the models [2]. There have been many solutions to address the problem through Parameter-Efficient Fine-Tuning (PEFT), one of which is Low-Rank Adaptation (LoRA). LoRA showcases a promising performance on many tasks such as computer vision and NLP.

2 Related Work

Prior work such as FedIT [1], applies FedAvg to the LoRA adapters independently, which [2] demonstrated that it is mathematically inaccurate: averaging A and B separately introduce mathematical errors to the global LoRA, termed aggregation noise, and cannot combine adapters of different ranks. The proposed method, FLoRA, replaces averaging with stacking-based aggregation (Equation (1)): the local B_k are concatenated in column-wise and the weighted A_k row-wise, so that the product equals the exact weighted sum of local updates.

$$\sum_{k=0}^{K} B_k A_k = (B_0 \oplus \cdots \oplus B_K)(A_0 \oplus \cdots \oplus A_K) \tag{1}$$

Because the ranks are concatenated, clients can use heterogenous ranks r_k without padding. The aggregated update is merged into the frozen base weights W', and clients reinitialize their adapters each round. To limit what the shared stack reveals, local modules are split into rank-1 components and shuffled before stacking. Experiments with TinyLlama, Llama, and Llama2 on Databricks-dolly-15k instruction, Alpaca, Wizard, ShareGPT, MMLU, MT-bench, across 10 Non-Independent and Identically Distributed (non-IID) clients, showed that FLoRA outperforms full fine-tuning and slightly higher than FedIT.

[3] target system heterogeneity, where resource-constrained clients cannot train high-rank adapters. The framework build on HetLoRA, whose zero-padding strategy halves the parameters in the extra dimensions of higher-rank adapters during averaging, and propose two variants. HeLoRA-Pad replaces zero-padding with context-based padding, filling missing rows or columns from statistics of the existing matrix (maximum, mean, or a copy of the adjacent row or column), and aggregates with rank-based adaptive weights:

$$W_{avg} = \frac{\sum_{i=1}^{m} n_i r_i W_i}{\sum_{i=1}^{m} n_i r_i}, \tag{2}$$

before truncating the results back to each client's rank. HeLoRA-Pad averages the adapter metrics element-wise. HeLoRA-KD avoids parameter aggregation entirely: the

server evaluates each client's model on a public dataset, builds a consensus from the $n_i r_i$-weighted logits, and updates every model by deep mutual learning, resulting in a personalized model without a global model. Experiments use DistilBERT and ViT-B/16 on IMDB, SST-2, and CIFAR-10/100, with 20 clients in Independent and Identically Distributed (IID) and non-IID scenarios, and ranks drawn from $\{4, 8, 16, 32\}$, results report average gains of 2.3% for HeLoRA-KD over HetLoRA.

Under Differential Privacy (DP), clients in federated LoRA inject various noise levels according to their privacy preferences, and noisy updates may degrade the global model performance. Existing incentive mechanisms counter this through monetary payments, a trusted third-party, or Shapely-value pricing, which is impractical at LLM scale. Instead, [4] use the aggregation as an incentive. The server estimates each client's noise scale $\hat{\sigma}_i$ via leave-one-out PCA: the residual energy of client i's LoRA update outside the subspace spanned by the other client's updates serve as the noise estimate, thus no self-reported preferences are required. Then updates are aggregated with inverse-noise weights:

$$\Delta W_g^t = \sum_{i=1}^{N} w_i^t B_i^t A_i^t, \qquad w_i^t = \frac{1/(\hat{\sigma}_i^t + \tau)}{\sum_{j=1}^{N} 1/(\hat{\sigma}_j^t + \tau)}. \tag{3}$$

Clients select their noise levels with an upper-confidence-bound strategy that trades local accuracy against privacy, and experiments prove that the game admits a stationary Markov equilibrium. Experiments with TinyLlama and GPT2-Large on AGNews, DBpedia, and 20Newsgroups (10 clients, Dirichlet $\alpha = 0.3$) show that WinFLoRA exceeds FedIT in FLoRA in global accuracy by up to 50.66% and 52.58% at matched average noise, respectively.

[5]

3 Experimental Analysis

4 Critical Evaluation

5 Analysis

6 Conclusion

References

[1] J. Zhang et al., "Towards building the federatedgpt: Federated instruction tuning," in *ICASSP 2024 - 2024 IEEE International Conference on Acoustics, Speech and Signal*

Table 1: Comparative summary of the reviewed federated LoRA methods.

Method	Aggregated	Exact	Hetero	Privacy	Key strength	Key limitation
FLoRA [2]	A, B (stacked)	✓	Rank, data	Shuffling; DP claimed	Exact aggregation with heterogeneous ranks, no padding	Download grows with $\sum_k r_k$; evaluated over 1–3 rounds
HeLoRA [3]	Padded A, B averaged (Pad); logits (KD)	× / –	Rank, data	None; server holds client models and a public dataset	Supports heterogeneous ranks; KD preserves each client's adapter structure	Pad aggregation remains inexact; evaluated only on models up to 340M parameters and non-generative tasks
FedSA-LoRA [4]						
WinFLoRA [4]	A, B (product-level, inverse-noise weighted)	✓	Privacy level, data	DP noise; no formal (ε, δ) accounting	Noise estimated from updates; no payments or trusted third party	Classification only, models $\leq$1.1B; privacy measured as normalized noise scale

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[2] Z. Wang et al., "Flora: Federated fine-tuning large language models with heterogeneous low-rank adaptations," in *Advances in Neural Information Processing Systems 37*, 2024. DOI: 10.52202/079017-0708 arXiv: 2409.05976.

[3] B. Fan, X. Su, S. Tarkoma, and P. Hui, "Helora: Lora-heterogeneous federated fine-tuning for foundation models," *ACM Transactions on Internet Technology*, vol. 25, no. 2, pp. 1–22, May 2025, ISSN: 1533-5399, 1557-6051. DOI: 10.1145/3723877

[4] M. Kou et al., "Winflora: Incentivizing client-adaptive aggregation in federated lora under privacy heterogeneity," in *Proceedings of the ACM Web Conference 2026*, ser. WWW '26, New York, NY, USA: Association for Computing Machinery, Apr. 2026, pp. 5241–5252. DOI: 10.1145/3774904.3792295

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